



BobSim

SteadyStateEval Vehicle Characterization

Measured- a_y ramp-steer isolines with robust fits

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Run Notes

- Open-loop ramp-steer at 12.5, 15, 17.5, and 20 m/s
- steerStart = 2.0 s; trim all pre-ramp samples
- Measured a_y is the analysis axis; commanded a_y only sets the ramp
- lmeasured a_y < 1.0 m/s² is excluded; linear band = 1.0 to 4.0 m/s²
- Understeer gradient = mean $d(\text{delta_excess})/d(a_y)$ over the linear band
- Response curves use split Hampel-filtered spline fits; sensitivities trim the outer 20% of lmeasured a_y
- steerExcess = avgSteerAngle - wheelbase * curvature
- Tire/config: MF52, baseline_v3; Backend: OpenModelica

SteadyStateEval Summary 12.5 m/s

Metric	Value	Units
Measured a_y Range	$-17.18 \rightarrow 17.57$	m/s^2
Roadwheel Angle Gradient	0.01026	$\frac{\text{rad}}{\text{m/s}^2}$
	5.76	$\frac{\text{deg}}{g}$
Handwheel Angle Gradient	0.04223	$\frac{\text{rad}}{\text{m/s}^2}$
	23.73	$\frac{\text{deg}}{g}$
Sideslip Gradient	0.00211	$\frac{\text{rad}}{\text{m/s}^2}$
	1.18	$\frac{\text{deg}}{g}$
Understeer Gradient	0.00055	$\frac{\text{rad}}{\text{m/s}^2}$
	0.31	$\frac{\text{deg}}{g}$
Handwheel Understeer Gradient	0.04223	$\frac{\text{rad}}{\text{m/s}^2}$
	23.73	$\frac{\text{deg}}{g}$
Roll Gradient	1.024	$\frac{\text{deg}}{g}$
Handwheel Torque Min	-19.7	$\text{N} \cdot \text{m}$
Handwheel Torque Max	19.4	$\text{N} \cdot \text{m}$

SteadyStateEval Summary 15.0 m/s

Metric	Value	Units
Measured a_y Range	$-17.55 \rightarrow 18.25$	m/s^2
Roadwheel Angle Gradient	0.00752	$\frac{\text{rad}}{\text{m/s}^2}$
	4.22	$\frac{\text{deg}}{g}$
Handwheel Angle Gradient	0.03099	$\frac{\text{rad}}{\text{m/s}^2}$
	17.42	$\frac{\text{deg}}{g}$
Sideslip Gradient	0.00082	$\frac{\text{rad}}{\text{m/s}^2}$
	0.46	$\frac{\text{deg}}{g}$
Understeer Gradient	0.00069	$\frac{\text{rad}}{\text{m/s}^2}$
	0.39	$\frac{\text{deg}}{g}$
Handwheel Understeer Gradient	0.03099	$\frac{\text{rad}}{\text{m/s}^2}$
	17.42	$\frac{\text{deg}}{g}$
Roll Gradient	1.044	$\frac{\text{deg}}{g}$
Handwheel Torque Min	-18.9	$\text{N} \cdot \text{m}$
Handwheel Torque Max	18.6	$\text{N} \cdot \text{m}$

SteadyStateEval Summary 17.5 m/s

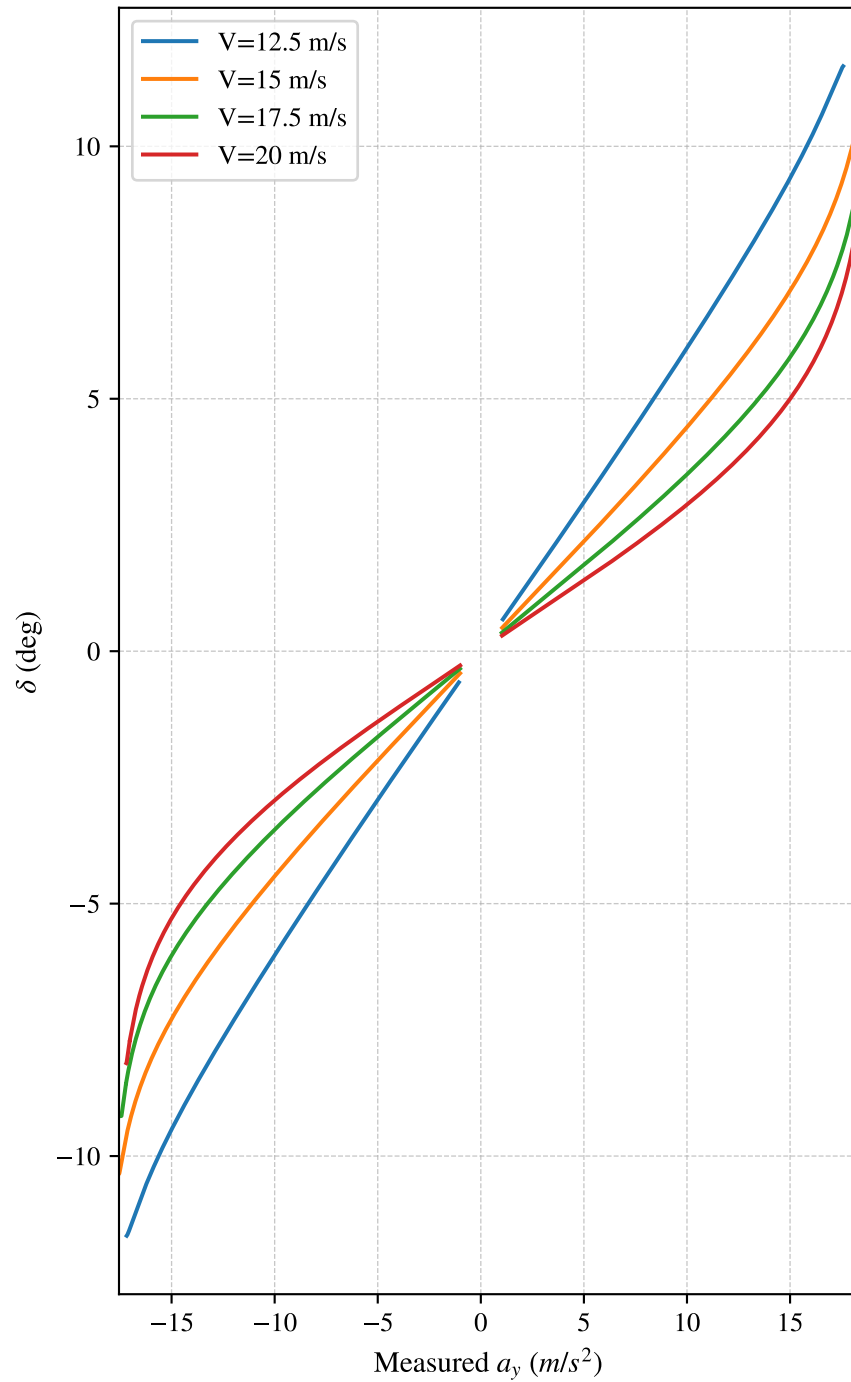
Metric	Value	Units
Measured a_y Range	$-17.44 \rightarrow 18.28$	m/s^2
Roadwheel Angle Gradient	0.00587	$\frac{\text{rad}}{\text{m/s}^2}$
	3.30	$\frac{\text{deg}}{g}$
Handwheel Angle Gradient	0.02421	$\frac{\text{rad}}{\text{m/s}^2}$
	13.61	$\frac{\text{deg}}{g}$
Sideslip Gradient	0.00008	$\frac{\text{rad}}{\text{m/s}^2}$
	0.04	$\frac{\text{deg}}{g}$
Understeer Gradient	0.00080	$\frac{\text{rad}}{\text{m/s}^2}$
	0.45	$\frac{\text{deg}}{g}$
Handwheel Understeer Gradient	0.02421	$\frac{\text{rad}}{\text{m/s}^2}$
	13.61	$\frac{\text{deg}}{g}$
Roll Gradient	1.067	$\frac{\text{deg}}{g}$
Handwheel Torque Min	-18.4	$\text{N} \cdot \text{m}$
Handwheel Torque Max	18.1	$\text{N} \cdot \text{m}$

SteadyStateEval Summary 20.0 m/s

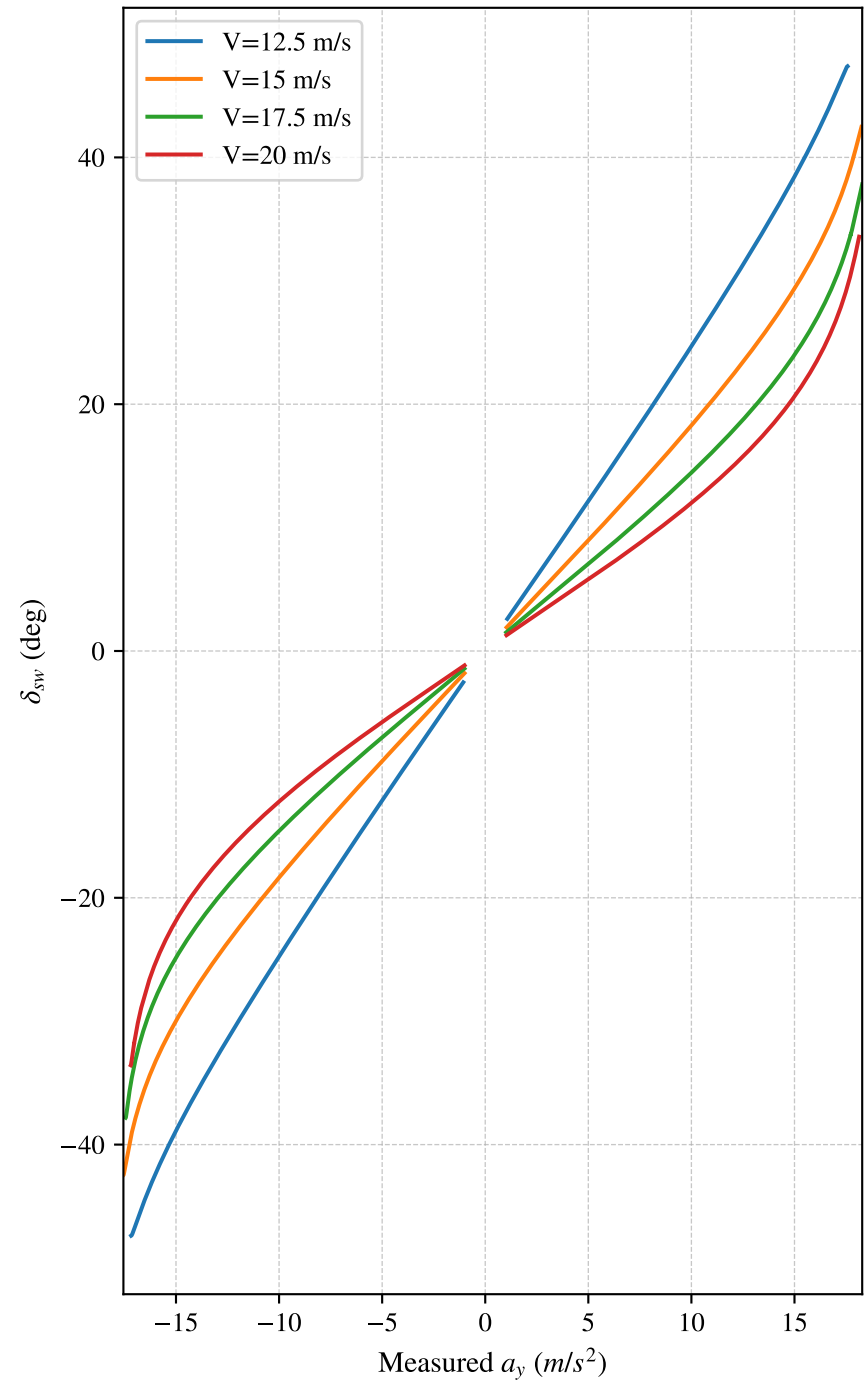
Metric	Value	Units
Measured a_y Range	$-17.19 \rightarrow 18.14$	m/s^2
Roadwheel Angle Gradient	0.00480	$\frac{\text{rad}}{\text{m/s}^2}$
	2.70	$\frac{\text{deg}}{g}$
Handwheel Angle Gradient	0.01983	$\frac{\text{rad}}{\text{m/s}^2}$
	11.15	$\frac{\text{deg}}{g}$
Sideslip Gradient	-0.00039	$\frac{\text{rad}}{\text{m/s}^2}$
	-0.22	$\frac{\text{deg}}{g}$
Understeer Gradient	0.00088	$\frac{\text{rad}}{\text{m/s}^2}$
	0.49	$\frac{\text{deg}}{g}$
Handwheel Understeer Gradient	0.01983	$\frac{\text{rad}}{\text{m/s}^2}$
	11.15	$\frac{\text{deg}}{g}$
Roll Gradient	1.090	$\frac{\text{deg}}{g}$
Handwheel Torque Min	-18.0	$\text{N} \cdot \text{m}$
Handwheel Torque Max	17.8	$\text{N} \cdot \text{m}$

Steering Angle Isolines vs Measured a_y

Roadwheel Angle δ

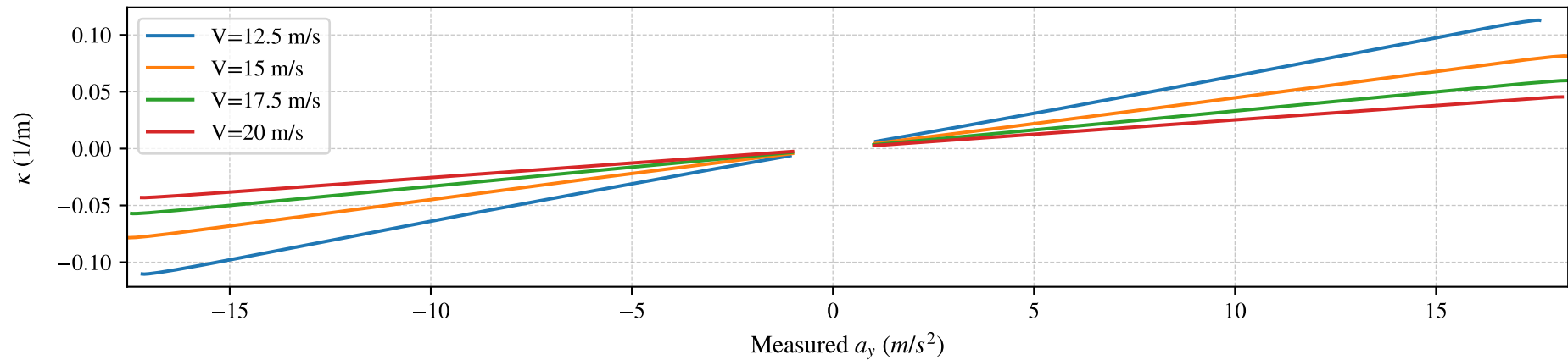


Handwheel Angle δ_{sw}

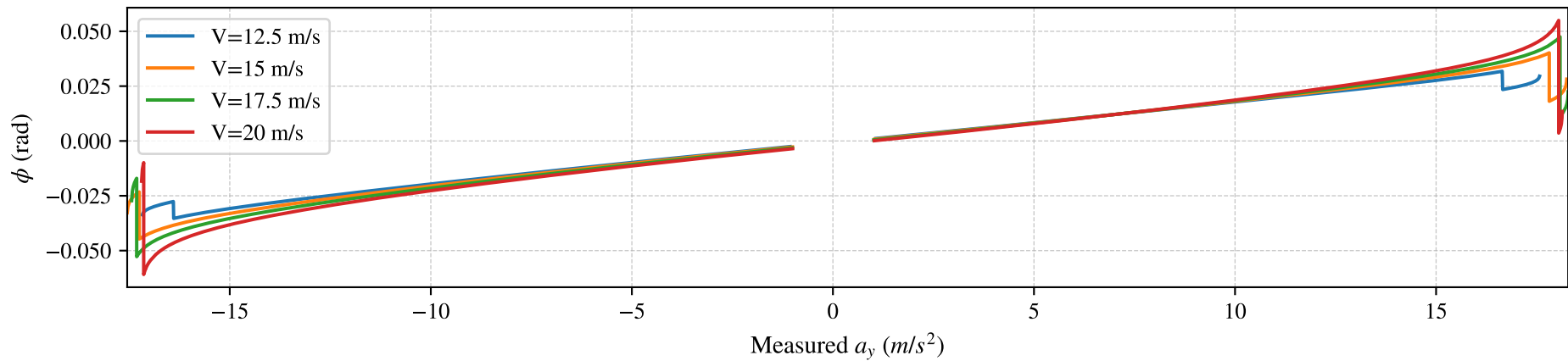


Vehicle State Isolines vs Measured a_y

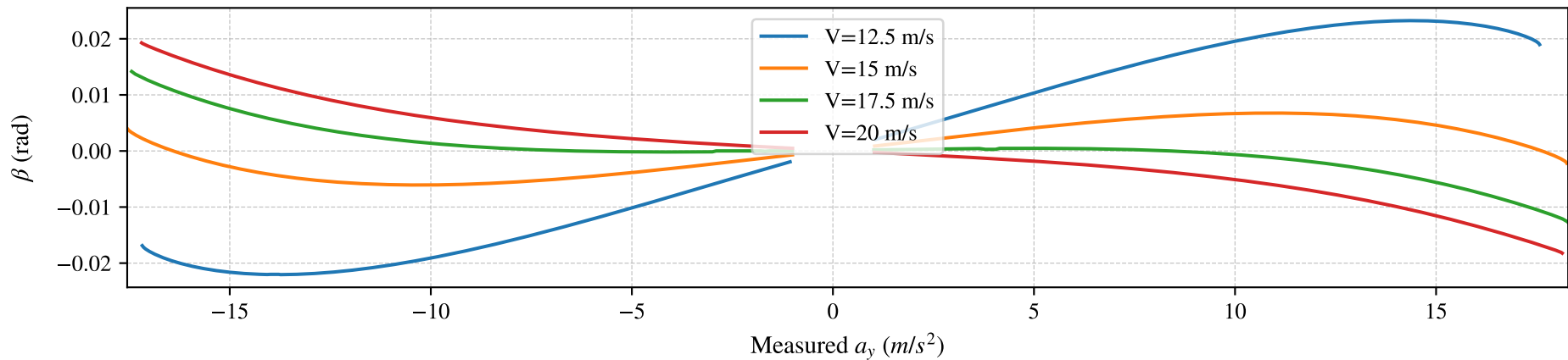
Curvature κ



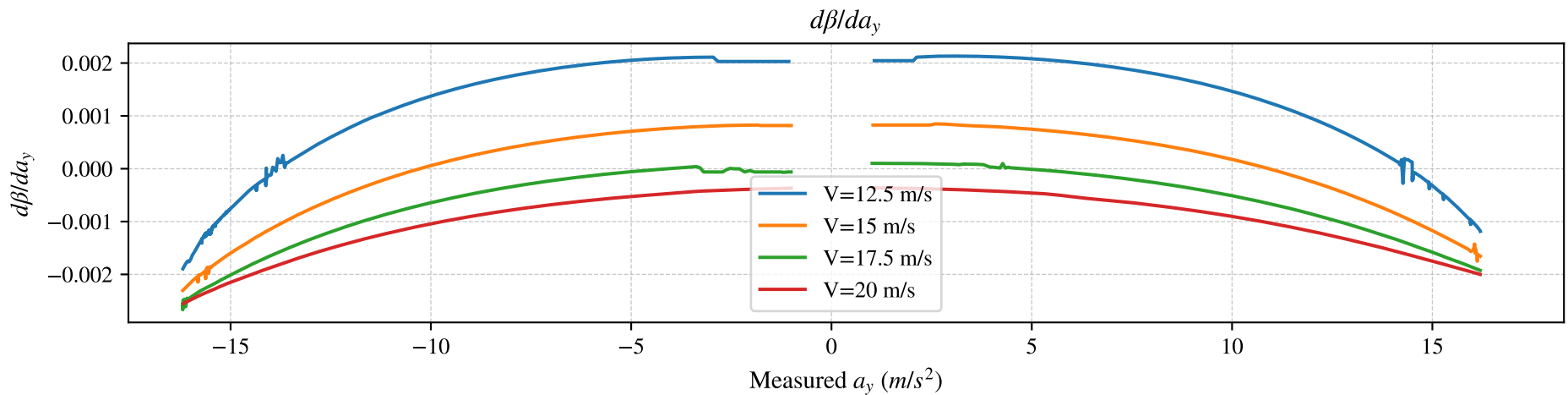
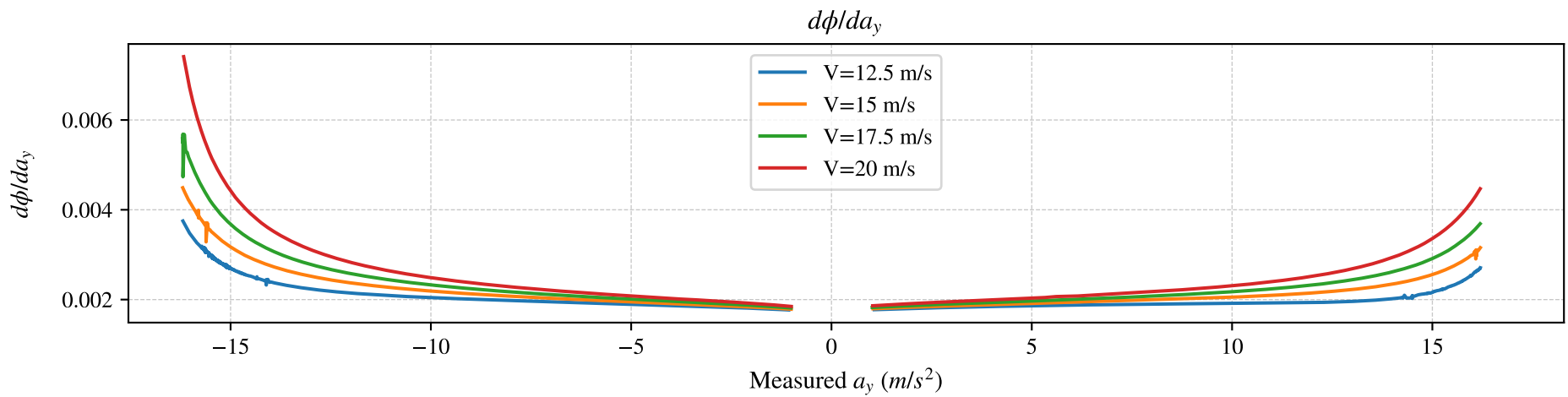
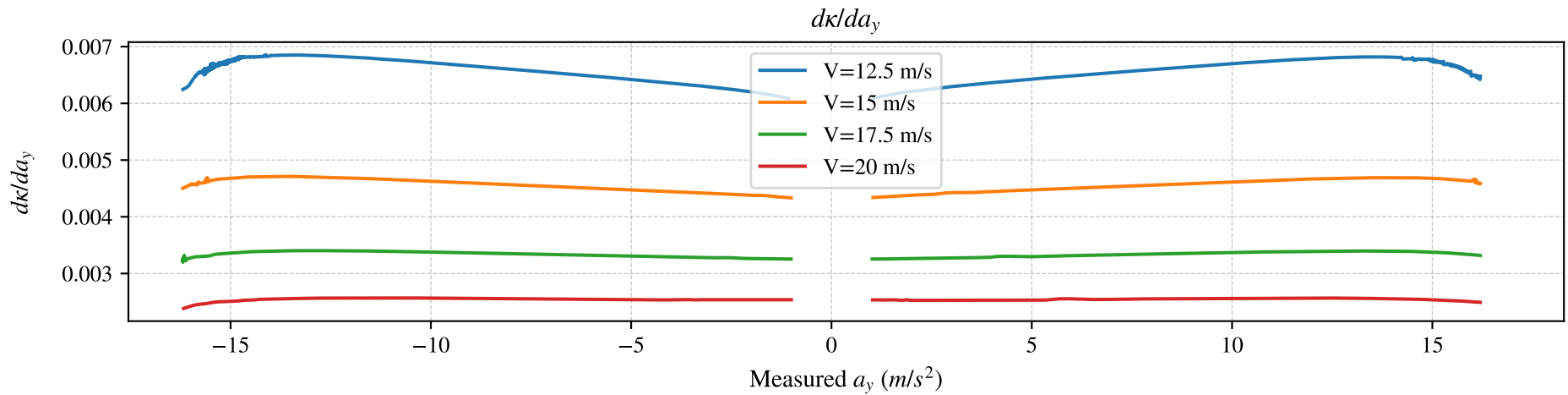
Roll Angle ϕ



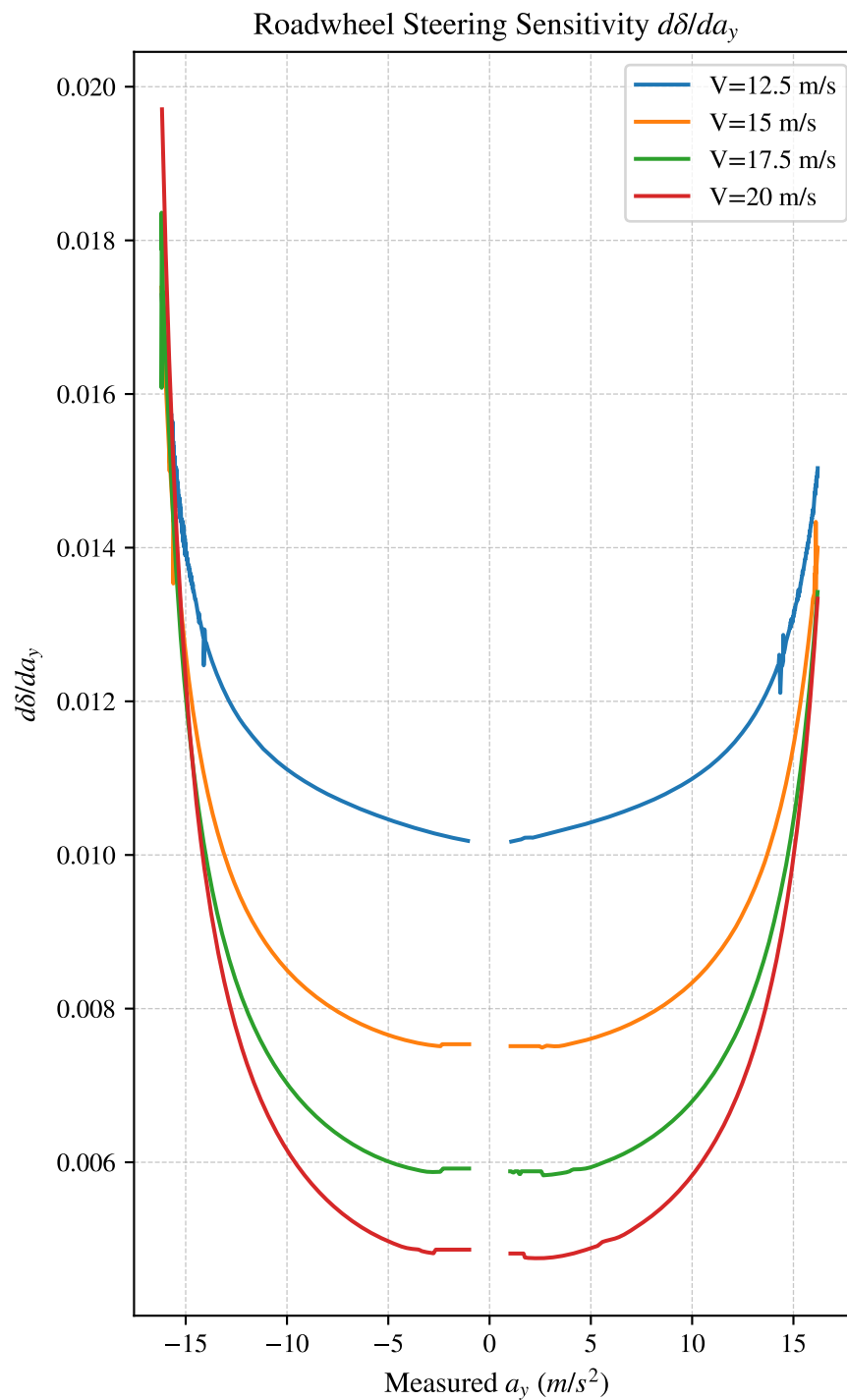
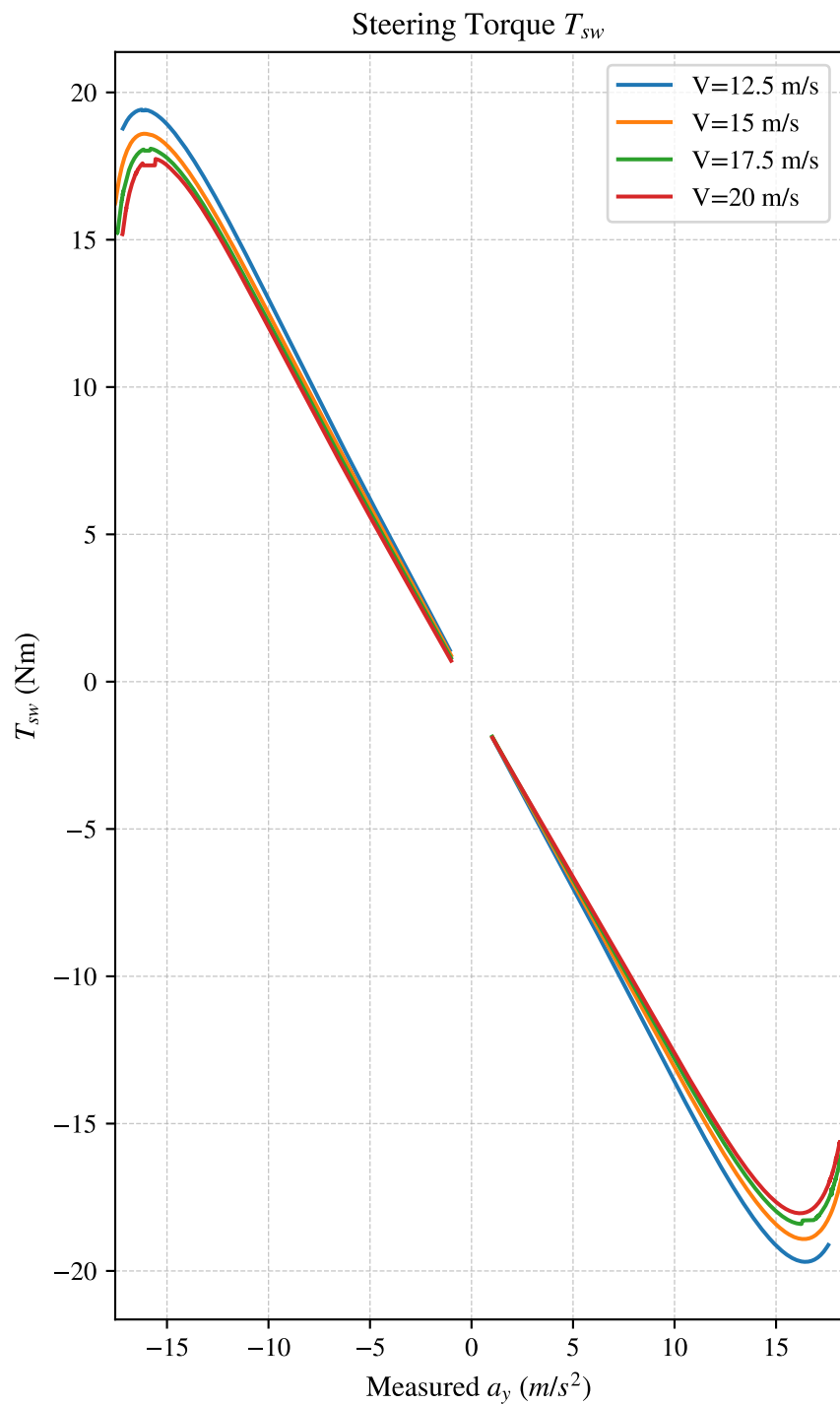
Sideslip β



Parametric Response Sensitivities vs Measured a_y

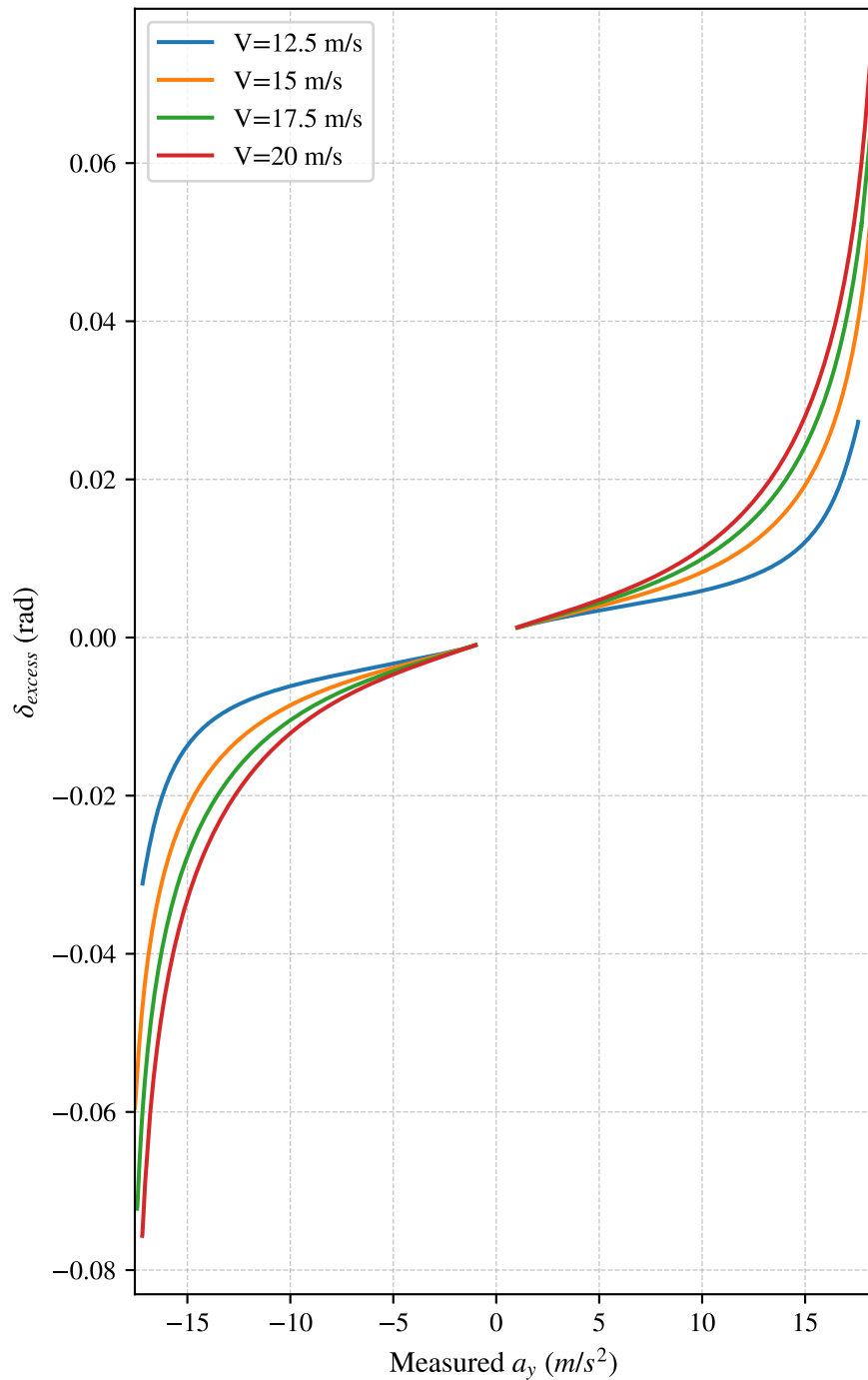


Steering Effort Isolines vs Measured a_y

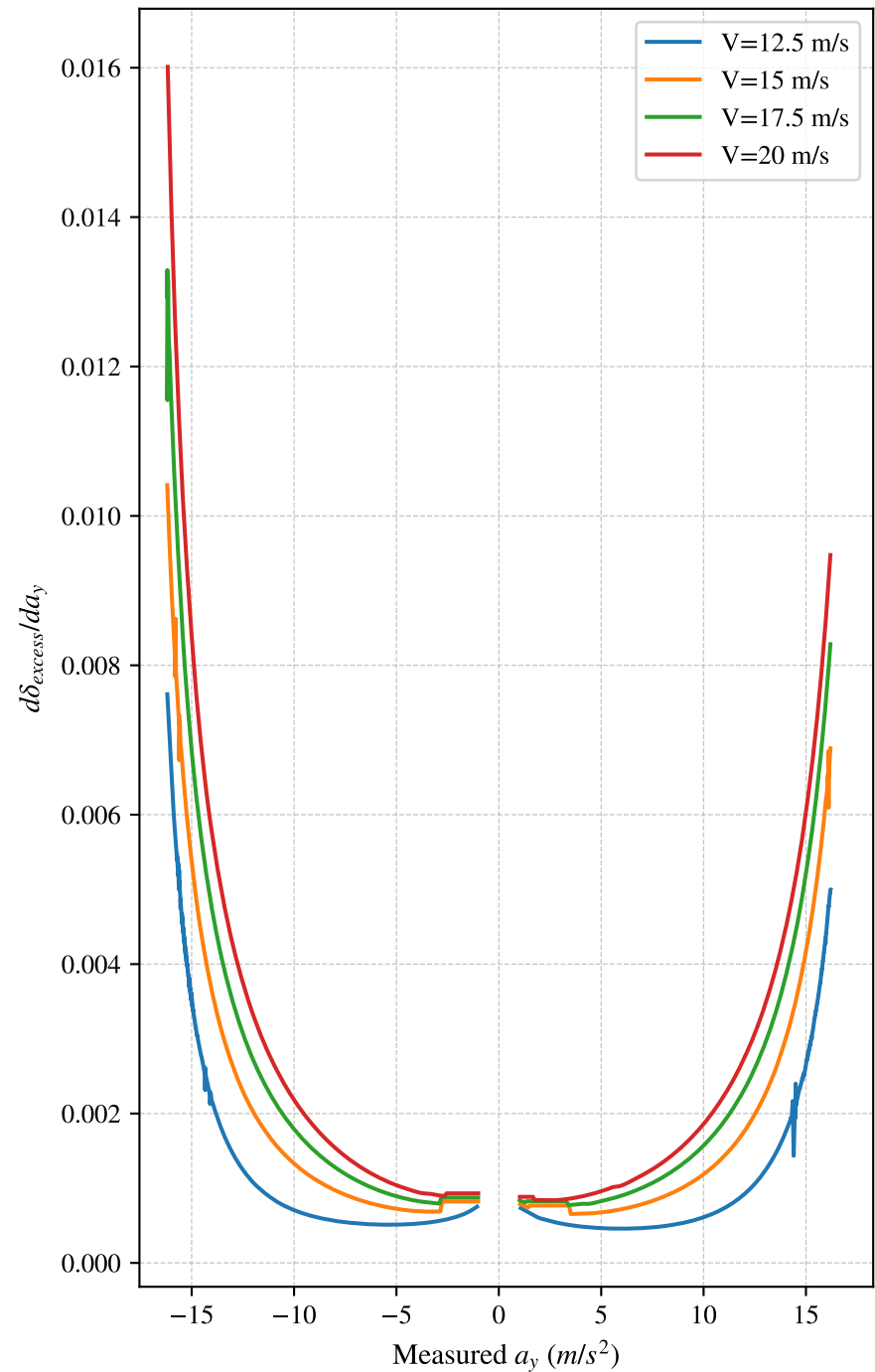


Excess Steer Isolines vs Measured a_y

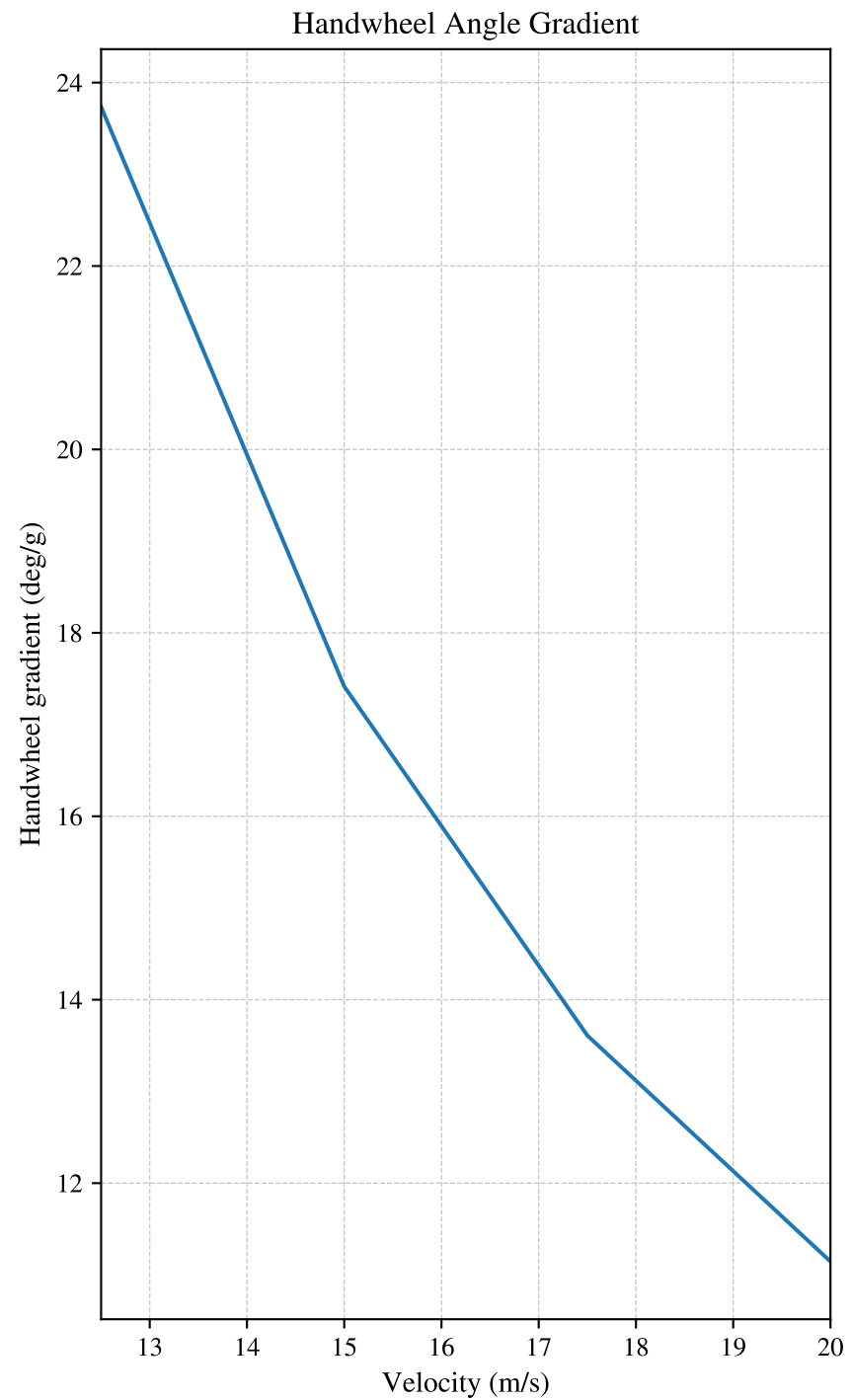
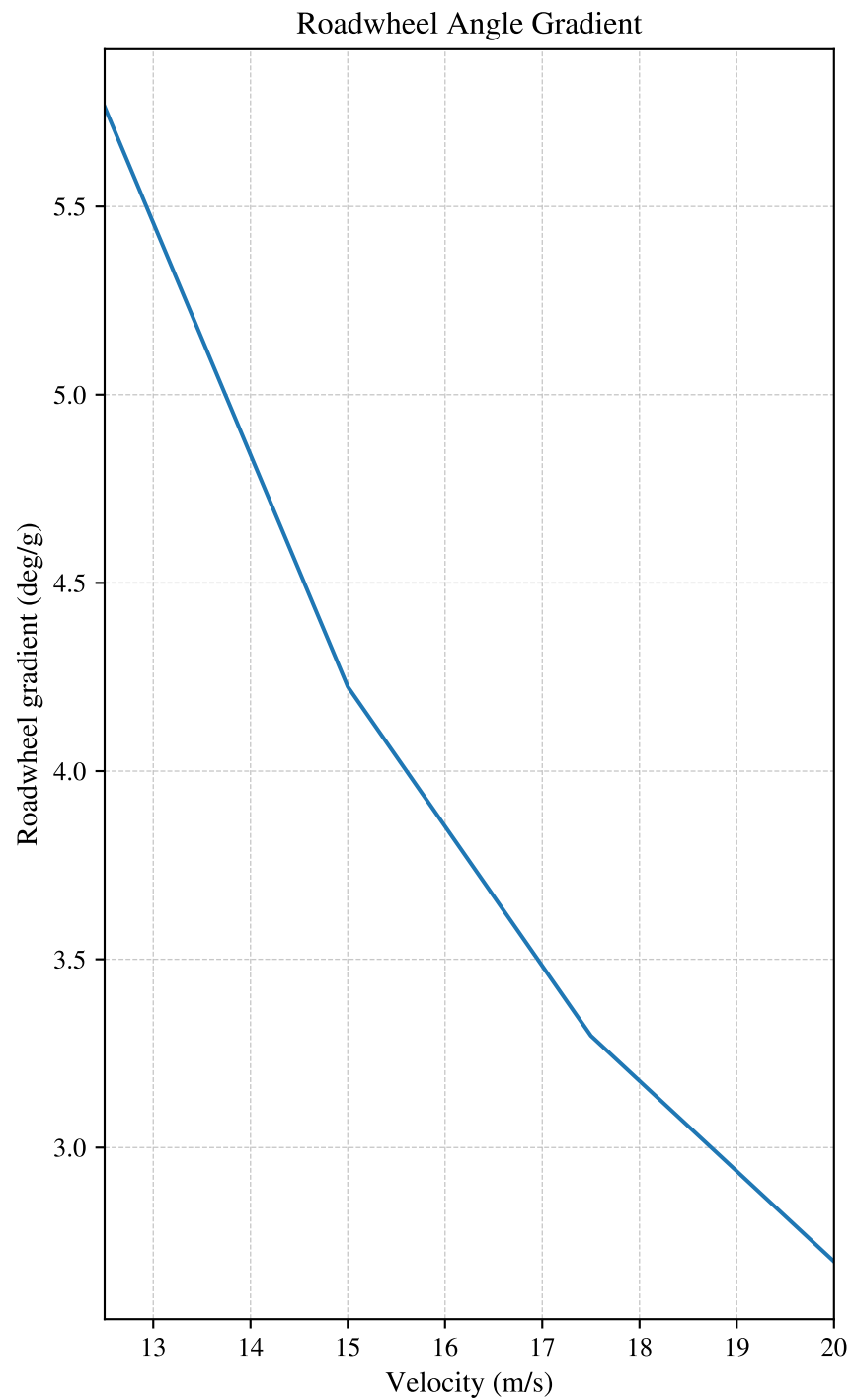
Steer Excess δ_{excess}



Understeer Gradient $d\delta_{excess}/da_y$

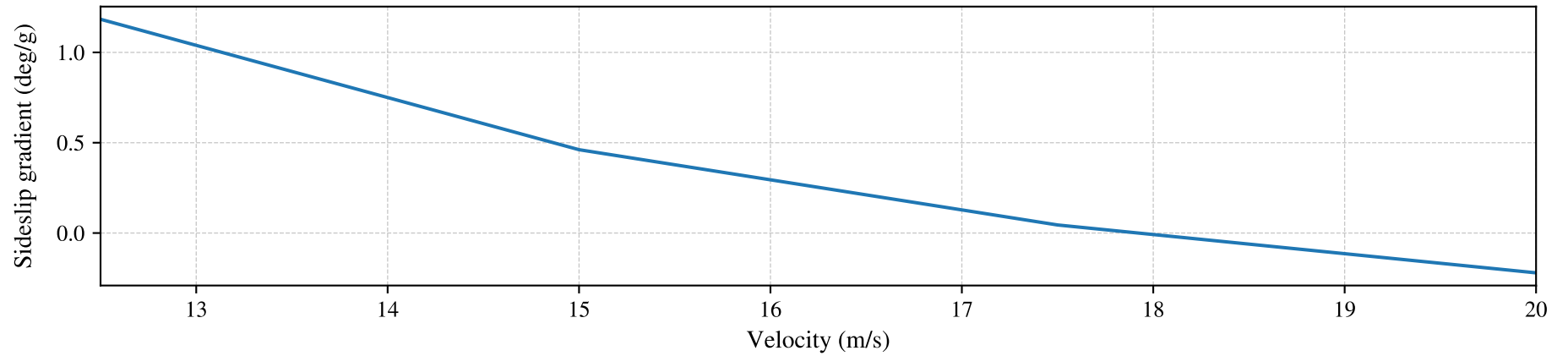


Angle Gradients vs Velocity

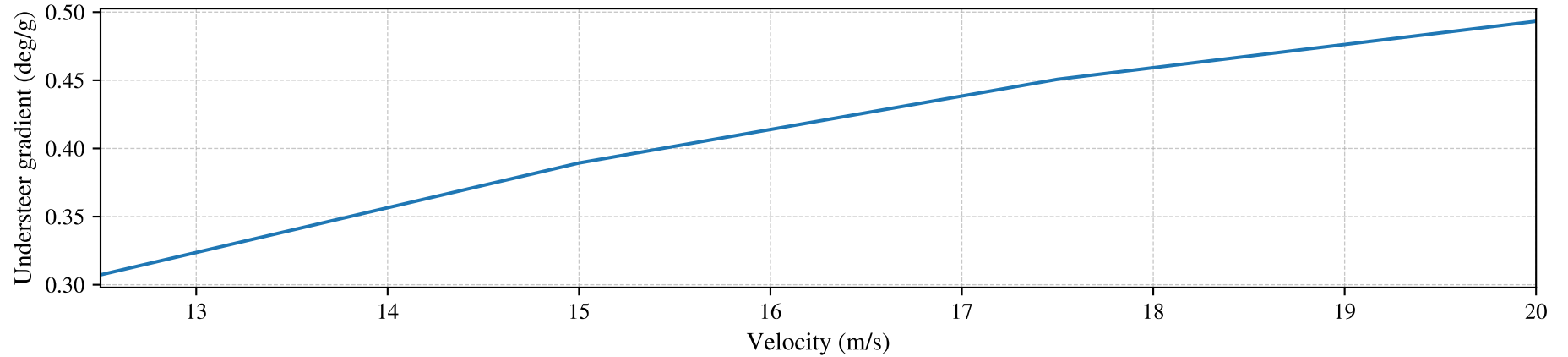


State Gradients vs Velocity

Sideslip Gradient



Understeer Gradient



Roll Gradient

